Syllabus
REAL VARIABLES I
Fall 2026

Item	Information
Course	REAL VARIABLES I MATH55003-01
Semester	Fall 2026
Instructor	Gennady Uraltsev
Email	uraltsev@uark.edu
Office	SCEN 421
Class meetings	Mo/We/Fr 9:40 - 10:30, CHPN 421
Office hours	Thursdays, 16:00–17:30 p.m., SCEN 421

Note: On weeks with departmental meetings, office hours begin at 4:30 p.m.

0.1 Course overview

University of Arkansas mathematics course listings
MATH 55003. Theory of Functions of a Real Variable I. 3 Hours.
Prerequisite: *MATH 45203 or MATH 52203; graduate standing in mathematics or statistics; or departmental consent.*

This course develops the foundations of Lebesgue measure and integration and the L^p spaces that underpin modern analysis, partial differential equations, probability, and harmonic analysis. We begin by identifying limitations of Riemann integration, then we construct measures - ways of describing how large a set is. We then develop measurable functions and use the notion of measure to construct the Lebesgue integral, describing how large functions are. Next, we prove the major limit theorems for the Lebesgue integral, then and study normed spaces of integrable functions. The final part of the course treats product measures and the Tonelli-Fubini theorems.

Unit	Main capability
From Riemann integration to measure	Explain why a more flexible integral is needed; work with countab
σ-algebras and measures	Construct and analyze measurable spaces, Borel sets, measures, ar
Measurable functions and Lebesgue integration	Verify measurability, approximate nonnegative functions by simple
L^p spaces	Work with almost-everywhere equivalence, Hölder and Minkowski
Product measures and iterated integration	Set up product measure spaces and apply Tonelli's and Fubini's th

0.1.1 Learning outcomes

By the end of the course, you should be able to:

1. State and use the definitions of outer measure, σ-algebra, measure, measurable function, almost-everywhere equality, and the Lebesgue integral.

2. Construct Lebesgue measure through outer measure and Carathéodory's criterion, and distinguish measurable from nonmeasurable sets.

3. Prove measurability and integrability claims using approximation by simple functions and superlevel-set methods.

4. Select and apply monotone convergence, Fatou's lemma, dominated convergence, Egorov's theorem, and uniform-integrability arguments.

5. Compare Riemann and Lebesgue integration.

6. Analyze L^p spaces using Hölder's and Minkowski's inequalities, completeness, density, convergence modes, and norm-testing arguments.

7. Use product measures and Tonelli-Fubini to justify iterated integration and interchange the order of integration.

8. Communicate graduate-level analysis in complete written proofs and in a prepared oral presentation, including motivation, examples, and counterexamples.

0.1.2 Materials

My lecture notes are our main guide through the course. They may also link to explanatory videos or other freely available sources.
Reference:
Luigi Ambrosio, Giuseppe Da Prato, and Andrea Mennucci, *Introduction to Measure Theory and Integration*, Publications of the Scuola Normale Superiore 10, Edizioni della Normale, 2011, doi:10.1007/978-88-7642-386-4.
Additional references are listed on the resources page.

0.2 Course organization

In class, I will introduce new topics, develop proofs, and work through examples and applications with you. I will assign weekly written homework. Fridays will usually emphasize problem solving, focused especially on appropriately marked problems from the homework. If you give a substantially correct presentation of a problem designated for discussion, I may propose a grade immediately; if you are happy with that grade, you will not need to include that problem in the written submission.
After most class meetings, online questions provide a quick review of basic skills. They are graded automatically, and most allow multiple attempts.
We will have two written exams: a two-hour, self-proctored midterm and a two-hour final.
We will also have two oral exams. Before each exam, I will publish a list of eligible topics. At the exam I will give you a choice between two topics and you will have 40 minutes to prepare, followed by about 30 minutes for your presentation and a mathematical discussion.

0.2.1 Attendance and participation

Attendance is strongly encouraged but not graded, and there is no absence penalty.
If you need to miss class, use the posted notes, readings, online questions, homework, and other resources to keep up independently. Please keep track of course material, announcements, deadlines, and assessments.
If your online work, homework, or exams suggest that your current study routine is not serving you, I may reach out with ideas such as regular attendance or office hours.
Please plan to take scheduled exams on time. Make-up exams are available only for excused conflicts or emergencies.

0.3 Academic integrity

We all share responsibility for creating a course environment built on honest work and learning. Please follow the University of Arkansas Academic Integrity Policy; suspected violations are handled through the University process. Specific expectations for collaboration and artificial intelligence appear on the assignments page.

0.4 Assessment

Component	Weight
Written hand-in homework	30%
Online questions	10%
Midterm	15%
Final	15%
Oral exam 1	15%
Oral exam 2	15%
Total	**100%**

Dropped assignments:

- The lowest 2 written-homework scores are dropped.
- The lowest 4 online-question scores are dropped. One additional online-question score is dropped if you complete all check-in work.

0.4.1 Weekly hand-in homework

Each homework is graded on a four-point scale. After feedback is returned, you have two days to resubmit; if the revision improves the solution, the higher score is recorded.
If you need more time, talk to me **before the deadline** so that we can work out an extension. We will handle unforeseen emergencies case by case.

0.4.2 Online questions

Online questions normally follow a lecture and are due by the start of our next class meeting. They focus on fluency with key vocabulary, examples, and basic deductions.

0.4.3 Written exams

There are two two-hour written exams: one midterm and one final. Each covers roughly half of the course topics. The midterm will be completed outside normal class hours; the final follows the Registrar's schedule.

0.4.4 Oral exams

For each oral exam, I will announce a topic list in advance. At the appointment you will receive two topics, choose one, and have 40 minutes to organize a board presentation. The presentation and follow-up discussion will last about 30 minutes. You should explain the motivation and precise statement, present the central proof, and discuss relevant examples, counterexamples, and limitations. Assessment emphasizes mathematical correctness, command of the ideas, and your ability to respond thoughtfully to questions.

0.4.5 Letter grades

The following scale is the tentative baseline. It is provided for reference only and can change through the semester.

Course percentage	Letter grade
88-100	A
76-87.99	B
64-75.99	C
51-63.99	D
Below 51	F

0.5 Communication, records, and course changes

Email is the best way to reach me with an individual question.
I will post course-wide announcements, grades, and updates in Blackboard, so please check your University email and Blackboard regularly.
If you notice a possible grading or recording error, please let me know within one week of the posted score.
I may revise the syllabus when needed. If a change materially affects grading, exams, deadlines, or course delivery, I will announce it in writing. The current version of this website is our course reference.

0.6 Other information

0.6.1 Accessibility and accommodations

If you have approved accommodations, please request your accommodation letter and contact me early so we can make arrangements in time. If you think you may need accommodations but have not yet set them up, the Center for Educational Access can help. Temporary conditions can affect access too, so please reach out promptly when they arise.

0.6.2 Religious observances, University activities, and other excusable conflicts

Please email me as early as possible about religious observances, required University activities, military service, jury duty, documented illness, a family crisis, or another excusable conflict. For an exam conflict you can anticipate, at least one week's notice is especially helpful. We will follow applicable University policies and deadlines.
If you have three or more final exams on one day, you may be able to reschedule one under Academic Policy 1500.20. If it involves this course, please make your request before the Registrar's withdrawal deadline.

0.6.3 Inclement weather and emergencies

We will follow University closure and inclement-weather decisions. If a closure prevents an exam, I will announce a new date through Blackboard and the course website. Please do not travel when conditions are unsafe.

- University emergency information
- RazALERT emergency notifications
- Academic Policy 1858.10: inclement weather

0.6.4 Student support

- Mathematics Resource and Teaching Center
- Student Success
- Counseling and Psychological Services
- U of A Cares

Additional links are collected on the resources page.